

Li Wang

PhD. Student

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🚩 Research Field: Computer Graphics, Human Computer Interaction

Education

Master and PhD, <i>Software Engineering, Department of Intelligence and Computing, Tianjin University</i>	2020 – 2026
Bachelor, <i>Measuring and Controlling Technologies and Instruments, School of Precision Instrument and Opto-electronics Engineering, Tianjin University</i>	2016 – 2020

Profile

- **Main research topic:** Material **appearance modeling** and **3D reconstruction/generation** in the field of computer graphics.
- **Main achievements:** Using smartphones and smartpads to complete **high-quality material appearance reconstruction** and geometric modeling through **lightweight acquisition**.
- **Main technology stack:** PBR rendering technology, deep learning model (including Diffusion large model), neural implicit field (NeRF)
- **Other research experience:** Using a combination of **multiple devices** such as robotic arms, leap motion, and handwriting tablets to realize experiential teaching system research

Selected Publications

EBREnv: SVBRDF Estimation in Uncontrolled Environment Lighting via Exemplar-Based Representation
SIGGRAPH Asia 2025

Li Wang, Jiajun Zhao, Lianghao Zhang, Fangzhou Gao, Jiawan Zhang*.

NFPLight: Deep SVBRDF Estimation via the Combination of Near and Far Field Point Lighting
SIGGRAPH Asia 2024 and ACM Trans. Graph.

Li Wang, Lianghao Zhang, Fangzhou Gao, Yuzhen Kang, and Jiawan Zhang*

DeepBasis: Hand-Held Single-Image SVBRDF Capture via Two-Level Basis Material Model
SIGGRAPH Asia 2023

Li Wang, Lianghao Zhang, Fangzhou Gao, and Jiawan Zhang*

MakeBronze: An interactive system to promote Chinese bronze culture in children through hands-on experience with lost-wax casting

International Journal of Human-Computer Studies, 2024. **(First Author among students)**
Minjing Yu, **Li Wang**, Mingxu Cai, Mengrui Zhang, Chun Yu, Xing-Dong Yang, Jiawan Zhang*

Deep SVBRDF Estimation from Single Image under Learned Planar Lighting
SIGGRAPH 2023

Lianghao Zhang, Fangzhou Gao, **Li Wang**, Minjing Yu, Jiamin Cheng, and Jiawan Zhang*

Transparent Object Reconstruction via Implicit Differentiable Refraction Rendering
SIGGRAPH Asia 2023

Fangzhou Gao, Lianghao Zhang, **Li Wang**, Jiamin Cheng, and Jiawan Zhang*

Single-image SVBRDF estimation with auto-adaptive high-frequency feature extraction
Computers & Graphics

Jiamin Cheng, **Li Wang**, Lianghao Zhang, Fangzhou Gao, Jiawan Zhang*

Projects

Internship – Tencent

04/2025 – 10/2025

- Conduct research on **3D PBR material generation**, focusing on **supervised fine-tuning (SFT)** of text-to-image models (e.g., Stable Diffusion), and integrating **physics-based priors** into the generation process.
- Responsible for generating large-scale **3D material datasets** to address data scarcity in PBR material modeling.
- Participated in SFT training of text-to-image foundation models, including **FLUX** and **SANA**.

Material appearance and geometric reconstruction based on lightweight equipment acquisition

2022 – present

- Responsible for the research on **material appearance reconstruction** using **smartphones** as acquisition devices:
 - Facing the problem of high **ambiguity** under single measurement values, a method combining the **basis material model with deep learning** is proposed, and the spatial relationship is successfully used to improve the reconstruction quality.
 - Facing the contradiction between **acquisition efficiency** and **acquisition quality**, an acquisition scheme based on the combination of **near-field and far-field shooting** is proposed, which achieves high-quality results comparable to the **original method of 20 images** input using **only 2 images** as input.
- Participated in the reconstruction of material appearance using **smart pad** as acquisition devices:
 - Based on **LTC real-time planar light source rendering** technology, a single surface light source material acquisition that can learn light source patterns is proposed.
- Participated in the research of transparent object reconstruction using only **smartphones** as acquisition devices:
 - Based on the **neural SDF field technology**, proposed 2D reprojected object contour extraction, thereby realizing fully automatic transparent object reconstruction without calibration.

Cultural heritage protection based on multi-device integration

2020 – 2021

Responsible for developing a safe, convenient, experiential teaching system for youths around **lost wax casting**, a representative production process of bronze ware, to promote the dissemination of Chinese bronze culture among young people. This system includes a **robotic arm**, leap motion, projector, drawing tablet and etc.

More Experience

Academic Reviewer

Reviewed SIGGRAPH, SIGGRAPH Asia, Chinagraph, IEEE Vis, etc.

Management

Served as a tutor for the first class of the undergraduate, responsible for the daily management of students, for a term of one year

Fund Application Writing

National Natural Science Foundation 2024 Project - Material Capture Topic; National Key R&D Program 2023 - Robotics Topic

Teaching

Teaching Matlab, analog circuits, single-chip computers, etc., one-year term